Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.53)

Metric

$$ds^2 = -e^{(2\,\nu(r,t))}\mathrm{d}t \otimes \mathrm{d}t + e^{(2\,\lambda(r,t))}\mathrm{d}r \otimes \mathrm{d}r + \left(-\frac{Y\left(r,t\right)^2}{\rho^2-1}\right)\mathrm{d}\rho \otimes \mathrm{d}\rho + \rho^2 Y\left(r,t\right)^2\,\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$(x^\mu) = (t, r, \rho, \phi)$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}e^{(\nu(r,t))}\right)\,dt + \left(-\sqrt{\frac{1}{2}}e^{(\lambda(r,t))}\right)\,dr$$

$$n = \left(\sqrt{\frac{1}{2}}e^{(\nu(r,t))}\right)\,dt + \left(\sqrt{\frac{1}{2}}e^{(\lambda(r,t))}\right)\,dr$$

$$m = \left(\frac{\sqrt{\frac{1}{2}}Y\left(r,t\right)}{\sqrt{-\rho^2+1}}\right)\,d\rho + \left(i\,\sqrt{\frac{1}{2}}\rho Y\left(r,t\right)\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}}Y\left(r,t\right)}{\sqrt{-\rho^2+1}}\right)\,d\rho + \left(-i\,\sqrt{\frac{1}{2}}\rho Y\left(r,t\right)\right)\,d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}e^{(-\nu(r,t))}\right) dt + \left(-\frac{1}{2}\sqrt{2}e^{(-\lambda(r,t))}\right) dr$$

$$n = \left(-\frac{1}{2}\sqrt{2}e^{(-\nu(r,t))}\right) dt + \left(\frac{1}{2}\sqrt{2}e^{(-\lambda(r,t))}\right) dr$$

$$m = \left(\frac{\sqrt{2}\sqrt{\rho+1}\sqrt{\rho-1}}{2\,Y\,(r,t)}\right) d\rho + \left(\frac{i\,\sqrt{2}}{2\,\rho Y\,(r,t)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}\sqrt{\rho+1}\sqrt{\rho-1}}{2\,Y\,(r,t)}\right) d\rho + \left(-\frac{i\,\sqrt{2}}{2\,\rho Y\,(r,t)}\right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}e^{(-\nu(r,t))}\,\partial_t - \frac{1}{2}\sqrt{2}e^{(-\lambda(r,t))}\,\partial_r$$

$$\Delta = -\frac{1}{2}\sqrt{2}e^{(-\nu(r,t))}\,\partial_t + \frac{1}{2}\sqrt{2}e^{(-\lambda(r,t))}\,\partial_r$$

$$\delta = \frac{\sqrt{2}\sqrt{\rho+1}\sqrt{\rho-1}}{2\,Y\,(r,t)}\,\partial_\rho + \frac{i\,\sqrt{2}}{2\,\rho Y\,(r,t)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{\rho+1}\sqrt{\rho-1}}{2\,Y\,(r,t)}\,\partial_\rho - \frac{i\,\sqrt{2}}{2\,\rho Y\,(r,t)}\,\partial_\phi$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = -\frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{2\,Y(r,t)} - \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{2\,Y(r,t)}$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = -\frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{2\,Y(r,t)} + \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{2\,Y(r,t)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t) - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)$$

$$\gamma = \frac{1}{4}\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t) - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)$$

Ricci Tensor Components

$$\Phi_{00} = \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\,\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\,\frac{\partial}{\partial t}\lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\,\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\,\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)}$$
$$+ \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\,\frac{\partial}{\partial r}\nu(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\,\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{2\,Y(r,t)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)}{Y(r,t)} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right)^2 + \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right) - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\nu\left(r,t\right)^2 - \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right)$$
$$+ \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}\lambda\left(r,t\right) - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}\nu\left(r,t\right) + \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right)^2}{4\,Y\left(r,t\right)^2} - \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right)^2}{4\,Y\left(r,t\right)^2} - \frac{1}{4\,Y\left(r,t\right)^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{8}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right)^2 - \frac{1}{8}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right) + \frac{1}{8}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\nu\left(r,t\right)^2 + \frac{1}{8}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right)$$
$$- \frac{1}{8}\, e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}\lambda\left(r,t\right) + \frac{1}{8}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}\nu\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right)^2}{8\,Y\left(r,t\right)^2} + \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right)^2}{8\,Y\left(r,t\right)^2} + \frac{1}{8\,Y\left(r,t\right)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right)^2 + \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right) - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\nu\left(r,t\right)^2 - \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right)$$
$$+ \frac{1}{4}\, e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}\lambda\left(r,t\right) - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}\nu\left(r,t\right) + \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right)^2}{4\,Y\left(r,t\right)^2} - \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right)^2}{4\,Y\left(r,t\right)^2} - \frac{1}{4\,Y\left(r,t\right)^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right) \frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right) \frac{\partial}{\partial t}\lambda\left(r,t\right)}{Y\left(r,t\right)} + \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right) \frac{\partial}{\partial t}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} + \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right)}{2\,Y\left(r,t\right)}$$
$$- \frac{e^{(-\lambda(r,t)-\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right)}{Y\left(r,t\right)} + \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}Y\left(r,t\right)}{2\,Y\left(r,t\right)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\, \frac{\partial^2}{\partial r \partial t}Y\left(r,t\right)}{Y\left(r,t\right)} - \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}Y\left(r,t\right)}{2\,Y\left(r,t\right)}$$

$$\Lambda = \frac{1}{12}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right)^2 + \frac{1}{12}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\lambda\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right) - \frac{1}{12}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}\nu\left(r,t\right)^2 - \frac{1}{12}\, e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}\lambda\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right) + \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right) \frac{\partial}{\partial r}\lambda\left(r,t\right)}{6\,Y\left(r,t\right)}$$
$$+ \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right) \frac{\partial}{\partial t}\lambda\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{1}{12}\, e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}\lambda\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right) \frac{\partial}{\partial r}\nu\left(r,t\right)}{6\,Y\left(r,t\right)} - \frac{1}{12}\, e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}\nu\left(r,t\right)$$
$$- \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right) \frac{\partial}{\partial t}\nu\left(r,t\right)}{6\,Y\left(r,t\right)} - \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial}{\partial r}Y\left(r,t\right)^2}{12\,Y\left(r,t\right)^2} - \frac{e^{(-2\,\lambda(r,t))}\, \frac{\partial^2}{(\partial r)^2}Y\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial}{\partial t}Y\left(r,t\right)^2}{12\,Y\left(r,t\right)^2} + \frac{e^{(-2\,\nu(r,t))}\, \frac{\partial^2}{(\partial t)^2}Y\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{1}{12\,Y\left(r,t\right)^2}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$\Psi_2 = 0$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

$$NP1 = \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)}$$
$$+ \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{2\,Y(r,t)}$$
$$- \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)}{Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{Y(r,t)^2} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)}$$

$$NP2 = \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)}$$
$$+ \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{2\,Y(r,t)}$$
$$- \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)}{Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{Y(r,t)^2} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)}$$

$$NP6 = \frac{7}{24}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2 + \frac{7}{24}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{7}{24}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{7}{24}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)$$
$$- \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{6\,Y(r,t)} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{6\,Y(r,t)} + \frac{7}{24}\,e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t) + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6\,Y(r,t)} - \frac{7}{24}\,e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\nu(r,t)$$
$$+ \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{6\,Y(r,t)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{24\,Y(r,t)^2} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{6\,Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{24\,Y(r,t)^2} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{6\,Y(r,t)} + \frac{1}{24\,Y(r,t)^2}$$

$$
\begin{aligned}
NP7 = {} & \frac{1}{4}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)^{2} + \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right) - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu\left(r,t\right)^{2} - \frac{1}{4}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right) + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} \\
& + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} + \frac{1}{4}\, e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}\lambda\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{1}{4}\, e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}\nu\left(r,t\right) \\
& - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right)}{2\,Y\left(r,t\right)} + \frac{3\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)^{2}}{4\,Y\left(r,t\right)^{2}} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{3\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)^{2}}{4\,Y\left(r,t\right)^{2}} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}Y\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{1}{4\,Y\left(r,t\right)^{2}}
\end{aligned}
$$

$$
\begin{aligned}
NP8 = {} & \frac{1}{6}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)^{2} + \frac{1}{6}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right) - \frac{1}{6}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu\left(r,t\right)^{2} - \frac{1}{6}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right) + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\lambda\left(r,t\right)}{3\,Y\left(r,t\right)} \\
& + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\lambda\left(r,t\right)}{3\,Y\left(r,t\right)} + \frac{1}{6}\, e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}\lambda\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right)}{3\,Y\left(r,t\right)} - \frac{1}{6}\, e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}\nu\left(r,t\right) \\
& - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right)}{3\,Y\left(r,t\right)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)^{2}}{3\,Y\left(r,t\right)^{2}} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(r,t\right)}{3\,Y\left(r,t\right)} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)^{2}}{3\,Y\left(r,t\right)^{2}} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}Y\left(r,t\right)}{3\,Y\left(r,t\right)} + \frac{1}{6\,Y\left(r,t\right)^{2}}
\end{aligned}
$$

$$
\begin{aligned}
NP10 = {} & -\frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial t}\lambda\left(r,t\right)}{Y\left(r,t\right)} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\lambda\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right)}{2\,Y\left(r,t\right)} \\
& + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right)}{Y\left(r,t\right)} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right)}{2\,Y\left(r,t\right)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)^{2}}{2\,Y\left(r,t\right)^{2}} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(r,t\right)}{2\,Y\left(r,t\right)} \\
& - \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial^{2}}{\partial r\partial t}Y\left(r,t\right)}{Y\left(r,t\right)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial t}Y\left(r,t\right)}{Y\left(r,t\right)^{2}} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)^{2}}{2\,Y\left(r,t\right)^{2}} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}Y\left(r,t\right)}{2\,Y\left(r,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
NP12 = {} & -\frac{1}{24}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)^{2} - \frac{1}{24}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right) + \frac{1}{24}\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu\left(r,t\right)^{2} + \frac{1}{24}\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right) \\
& + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\lambda\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\lambda\left(r,t\right)}{6\,Y\left(r,t\right)} - \frac{1}{24}\, e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}\lambda\left(r,t\right) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)\frac{\partial}{\partial r}\nu\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{1}{24}\, e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}\nu\left(r,t\right) \\
& - \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)\frac{\partial}{\partial t}\nu\left(r,t\right)}{6\,Y\left(r,t\right)} - \frac{17\, e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y\left(r,t\right)^{2}}{24\,Y\left(r,t\right)^{2}} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{17\, e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y\left(r,t\right)^{2}}{24\,Y\left(r,t\right)^{2}} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial^{2}}{(\partial t)^{2}}Y\left(r,t\right)}{6\,Y\left(r,t\right)} + \frac{5}{24\,Y\left(r,t\right)^{2}}
\end{aligned}
$$

$$NP14 = \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)}$$
$$- \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{2\,Y(r,t)}$$
$$+ \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)}{Y(r,t)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{Y(r,t)^2} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{2\,Y(r,t)^2} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)}$$

$$NP16 = \frac{1}{4}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2 + \frac{1}{4}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{4}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{4}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)}$$
$$+ \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)} + \frac{1}{4}\,e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)} - \frac{1}{4}\,e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\nu(r,t)$$
$$- \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)} + \frac{3\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{4\,Y(r,t)^2} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{2\,Y(r,t)} - \frac{3\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{4\,Y(r,t)^2} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)} - \frac{1}{4\,Y(r,t)^2}$$

$$NP17 = -\frac{1}{6}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2 - \frac{1}{6}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{6}\,e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)^2 + \frac{1}{6}\,e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{3\,Y(r,t)}$$
$$- \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{3\,Y(r,t)} - \frac{1}{6}\,e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t) + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{3\,Y(r,t)} + \frac{1}{6}\,e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\nu(r,t)$$
$$+ \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{3\,Y(r,t)} - \frac{e^{(-2\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2}{3\,Y(r,t)^2} + \frac{e^{(-2\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)}{3\,Y(r,t)} + \frac{e^{(-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2}{3\,Y(r,t)^2} - \frac{e^{(-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)}{3\,Y(r,t)} - \frac{1}{6\,Y(r,t)^2}$$

Equations are vanished.

Bianchi Identities

$$
\begin{aligned}
BI2 = & -\frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 + \frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) \\
& + \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2 + \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)} + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)} + \frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t) + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)} \\
& + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)} + \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)} - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)} \\
& + \frac{3}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) - \frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)} \\
& - \frac{3}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t) - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{4\,Y(r,t)^2} \\
& + \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t) + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8\,Y(r,t)} + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8\,Y(r,t)} + \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) \\
& + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{8\,Y(r,t)} - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{8\,Y(r,t)} - \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^3}{(\partial r)^3}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\nu(r,t) \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^3}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}Y(r,t)}{4\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{4\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{4\,Y(r,t)^2} \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)^2}{8\,Y(r,t)^3} + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^3}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{4\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{8\,Y(r,t)^3} + \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3}
\end{aligned}
$$

$$
\begin{aligned}
BI3 = & -\frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t)+\frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)-\frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) \\
& +\frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2-\frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2+\frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) \\
& +\frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t)-\frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2+\frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) \\
& +\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)}-\frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)}-\frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)+\frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)} \\
& -\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)}+\frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)-\frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& -\frac{1}{4}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)-\frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)}+\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)} \\
& +\frac{3}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)-\frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)-\frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& -\frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t)-\frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t)+\frac{1}{4}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& -\frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)-\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)}+\frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)} \\
& +\frac{3}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)+\frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t)+\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{4\,Y(r,t)^2} \\
& +\frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t)-\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2}-\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2} \\
& +\frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{4\,Y(r,t)^2}+\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8\,Y(r,t)}-\frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8\,Y(r,t)}-\frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) \\
& +\frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2}-\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2}-\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)^2} \\
& -\frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{8\,Y(r,t)}+\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{8\,Y(r,t)}-\frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^3}{(\partial r)^3}\nu(r,t)+\frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\nu(r,t) \\
& +\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)^2}-\frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^3}{8\,Y(r,t)^3}+\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}Y(r,t)}{4\,Y(r,t)^2} \\
& +\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3}-\frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{4\,Y(r,t)^2}+\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{4\,Y(r,t)^2} \\
& +\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)^2}{8\,Y(r,t)^3}-\frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^3}{8\,Y(r,t)^3}-\frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{4\,Y(r,t)^2}+\frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{8\,Y(r,t)^3}-\frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3}
\end{aligned}
$$

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\begin{aligned}
BI5 = & -\frac{1}{6}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{6}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{1}{6}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 + \frac{1}{6}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{6}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) \\
& + \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{1}{6}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2 + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)^2}{6\,Y(r,t)} \\
& + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{6\,Y(r,t)} + \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) + \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)} \\
& + \frac{3\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8\,Y(r,t)} + \frac{1}{6}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t) + \frac{5\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{24\,Y(r,t)} \\
& + \frac{3\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)} + \frac{1}{12}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6\,Y(r,t)} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6\,Y(r,t)} - \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& - \frac{3\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)} - \frac{3\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8\,Y(r,t)} + \frac{1}{4}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{6}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& - \frac{1}{6}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& - \frac{1}{6}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)} \\
& - \frac{5\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{24\,Y(r,t)} - \frac{1}{4}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t)}{6\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)^2}{6\,Y(r,t)} - \frac{1}{12}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t) + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\lambda(r,t)}{4\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{4\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{12\,Y(r,t)} - \frac{5\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{12\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\lambda(r,t)}{12\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}\lambda(r,t)}{12\,Y(r,t)} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{\partial r\partial t}\lambda(r,t)}{12\,Y(r,t)} + \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t) + \frac{2\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{3\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{6\,Y(r,t)} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12\,Y(r,t)} - \frac{5\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12\,Y(r,t)^2} \\
& - \frac{5\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{12\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12\,Y(r,t)} + \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8\,Y(r,t)} + \frac{7\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{24\,Y(r,t)} \\
& + \frac{1}{12}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) + \frac{5\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{12\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12\,Y(r,t)} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12\,Y(r,t)} \\
& + \frac{5\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12\,Y(r,t)^2} - \frac{2\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{3\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6\,Y(r,t)} \\
& \quad \frac{7\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{} \quad \frac{3\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{} \quad \frac{1}{} \quad \sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^3}{} \quad \frac{1}{} \quad \sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{}
\end{aligned}
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\begin{aligned}
BI6 = {} & \frac{1}{6}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{6}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{6}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 \\
& + \frac{1}{6}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{6}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) - \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t) \\
& + \frac{1}{6}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2 - \frac{\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)^2}{6Y(r,t)} + \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{6Y(r,t)} - \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) \\
& - \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8Y(r,t)} + \frac{3\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{8Y(r,t)} + \frac{1}{6}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t) - \frac{5\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{24Y(r,t)} \\
& + \frac{3\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{8Y(r,t)} - \frac{1}{12}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& - \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6Y(r,t)} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6Y(r,t)} + \frac{1}{6}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{3\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8Y(r,t)} - \frac{3\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{8Y(r,t)} - \frac{1}{4}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{6}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& + \frac{1}{6}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& - \frac{1}{6}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) + \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{8Y(r,t)} \\
& - \frac{5\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{24Y(r,t)} - \frac{1}{4}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t)}{6Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)^2}{6Y(r,t)} - \frac{1}{12}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t) - \frac{\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\lambda(r,t)}{4Y(r,t)^2} + \frac{\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{4Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{12Y(r,t)} - \frac{5\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{12Y(r,t)^2} + \frac{\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\lambda(r,t)}{12Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}\lambda(r,t)}{12Y(r,t)} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{\partial r\partial t}\lambda(r,t)}{12Y(r,t)} - \frac{1}{12}\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t) + \frac{2\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{3Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{6Y(r,t)} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12Y(r,t)} + \frac{5\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12Y(r,t)^2} \\
& - \frac{5\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{12Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{12Y(r,t)} - \frac{3\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{8Y(r,t)} + \frac{7\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{24Y(r,t)} \\
& + \frac{1}{12}\sqrt{2}e^{(-3\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) - \frac{5\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{12Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12Y(r,t)} + \frac{\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12Y(r,t)} \\
& + \frac{5\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{12Y(r,t)^2} + \frac{2\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{3Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{(\partial t)^2}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{6Y(r,t)} \\
& + \frac{7\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{24Y(r,t)} - \frac{3\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{8Y(r,t)} + \frac{1}{12}\sqrt{2}e^{(-3\lambda(r,t))}\frac{\partial^3}{(\partial r)^3}\nu(r,t) - \frac{1}{12}\sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\nu(r,t) \\
& \quad 11 \quad \sqrt{2}e^{(-2\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \quad \sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \quad \sqrt{2}e^{(-\lambda(r,t)-2\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)
\end{aligned}
$$

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\begin{aligned}
BI9 = & -\frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) \\
& + \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2 + \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{4\,Y(r,t)} + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{4\,Y(r,t)} + \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t) \\
& + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)} + \frac{1}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{4\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{4\,Y(r,t)} + \frac{3}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) - \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)} - \frac{3}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{1}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t) + \frac{3\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\lambda(r,t)}{8\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)^2} + \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t) + \frac{7\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{8\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{4\,Y(r,t)} \\
& + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{4\,Y(r,t)} + \frac{1}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)^2} \\
& - \frac{7\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{4\,Y(r,t)} - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{4\,Y(r,t)} - \frac{1}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^3}{(\partial r)^3}\nu(r,t) \\
& - \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\nu(r,t) + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)^2} - \frac{3\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)^2} \\
& + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^3}{8\,Y(r,t)^3} - \frac{3\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}Y(r,t)}{8\,Y(r,t)^2} - \frac{7\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}Y(r,t)}{8\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3} \\
& + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{2\,Y(r,t)^2} + \frac{7\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)^2}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^3}{8\,Y(r,t)^3} \\
& - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)^2} + \frac{3\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{8\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3}
\end{aligned}
$$

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\begin{aligned}
BI11 = {} & \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) - \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial r}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)^2 - \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)^2\frac{\partial}{\partial t}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)^2 - \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\lambda(r,t) \\
& - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{4\,Y(r,t)} + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)^2}{4\,Y(r,t)} + \frac{1}{8}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t) \\
& - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t)}{4\,Y(r,t)} - \frac{1}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^2}{(\partial r)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) \\
& + \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial r}\nu(r,t) + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{4\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)^2}{4\,Y(r,t)} - \frac{3}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) \\
& + \frac{1}{8}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t) + \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) + \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) \\
& - \frac{1}{8}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}\nu(r,t)\frac{\partial^2}{\partial r\partial t}\nu(r,t) + \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)} \\
& - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t)}{4\,Y(r,t)} - \frac{3}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^2}{(\partial t)^2}\lambda(r,t)\frac{\partial}{\partial t}\nu(r,t) - \frac{1}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t)\frac{\partial^2}{(\partial t)^2}\nu(r,t) - \frac{3\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\lambda(r,t)}{8\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\lambda(r,t)}{2\,Y(r,t)^2} - \frac{1}{16}\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^3}{\partial r(\partial t)^2}\lambda(r,t) + \frac{7\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{8\,Y(r,t)^2} \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\lambda(r,t)}{2\,Y(r,t)^2} - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{4\,Y(r,t)} \\
& + \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}\lambda(r,t)}{4\,Y(r,t)} + \frac{1}{16}\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial^3}{(\partial t)^3}\lambda(r,t) - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial r}\nu(r,t)}{2\,Y(r,t)^2} \\
& + \frac{7\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial r}\nu(r,t)}{8\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{4\,Y(r,t)} - \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial r)^2}\nu(r,t)}{4\,Y(r,t)} + \frac{1}{16}\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial^3}{(\partial r)^3}\nu(r,t) \\
& - \frac{1}{16}\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^3}{(\partial r)^2\partial t}\nu(r,t) - \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)\frac{\partial}{\partial t}\nu(r,t)}{2\,Y(r,t)^2} - \frac{3\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^2\frac{\partial}{\partial t}\nu(r,t)}{8\,Y(r,t)^2} \\
& - \frac{\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)^3}{8\,Y(r,t)^3} + \frac{3\sqrt{2}e^{(-3\,\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial r)^2}Y(r,t)}{8\,Y(r,t)^2} - \frac{7\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{\partial r\partial t}Y(r,t)}{8\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)^2\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3} \\
& + \frac{\sqrt{2}e^{(-2\,\lambda(r,t)-\nu(r,t))}\frac{\partial^2}{(\partial r)^2}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{2\,Y(r,t)^2} - \frac{7\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial^2}{\partial r\partial t}Y(r,t)\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial}{\partial t}Y(r,t)^2}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)^3}{8\,Y(r,t)^3} \\
& + \frac{\sqrt{2}e^{(-\lambda(r,t)-2\,\nu(r,t))}\frac{\partial}{\partial r}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{2\,Y(r,t)^2} + \frac{3\sqrt{2}e^{(-3\,\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)\frac{\partial^2}{(\partial t)^2}Y(r,t)}{8\,Y(r,t)^2} + \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{8\,Y(r,t)^3} - \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{8\,Y(r,t)^3}
\end{aligned}
$$

Equations are vanished.

Petrov Type

Type "O"